2026 Formula 1 Rear Wing Development

CFD Simulation Study

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ABSTRACT

This report presents the aerodynamic development of a 2026 Formula 1 rear wing using computational fluid dynamics (CFD) simulations. Seventeen high-downforce configurations were investigated using OpenFOAM, evaluating lift coefficient (C_L), drag coefficient (C_D), lift-to-drag ratio (L/D), and pitching moment (C_M). The study identifies optimal configurations balancing downforce generation with aerodynamic efficiency.

1 INTRODUCTION

The 2026 Formula 1 regulations introduce significant changes to rear wing design, emphasizing ground effect and active aerodynamics. This study focuses on CFD-based aerodynamic development of a rear wing configuration, systematically exploring design variations to maximize downforce while maintaining efficient drag characteristics.

The primary objectives of this study are:

- Evaluate aerodynamic performance across multiple wing configurations
- Identify optimal C_L/C_D ratio for maximum race pace performance
- Analyze the contribution of pressure and viscous forces to total drag
- Assess simulation convergence and numerical accuracy

2 METHODOLOGY

2.1 Simulation Setup

All simulations were conducted using OpenFOAM (version v2506) with the following configuration:

- **Solver**: simpleFoam (steady-state incompressible)
- **Turbulence Model**: k-ω SST
- **Mesh Type**: Cartesian mesh with refinement zones
- **Boundary Conditions**: Velocity inlet, pressure outlet, symmetry planes

2.2 Geometry

The rear wing geometry consists of a multi-element configuration stored in STL format (`rearWing.stl`). Each case variation modifies the wing flap angle, element spacing, and Gurney flap configurations to achieve different downforce levels.

2.3 Meshing and Grid Convergence Study

A mesh convergence study was performed using cfMesh to establish grid-independent solutions. Three mesh densities were evaluated: coarse (ID_04), medium (ID_02), and fine (ID_03). Table 1 summarizes the mesh configurations and key settings.

Table 1: *Mesh configuration summary for convergence study*

Parameter	Coarse (ID_04)	Medium (ID_02)	Fine (ID_03)
Number of cells	1,490,057	4,953,036	7,709,701
1st refinement box cell size	0.08	0.0156	0.01
2nd refinement box cell size	0.15	0.0625	0.0625
Local refinement (wall) cell size	0.035	0.005	0.0035
Edge refinement cell size	0.0015	0.0017	0.0017
Boundary layers	1	2	2
Max 1st layer thickness	0.01	0.008	0.008

Table 2 presents the aerodynamic coefficients for each mesh level, demonstrating consistent trends toward mesh independence.

The Grid Convergence Index (GCI) was computed using the formula:

$$GCI = F_s \cdot \frac{|f_2 - f_1|}{f_3 - f_1} \cdot \frac{1}{r^3 - 1} \tag{1}$$

Table 2: *Aerodynamic coefficients for mesh convergence study*

Mesh Level	$C_{D,mean}$	$C_{L,mean}$	Cells
Coarse (ID_04)	0.4585	-2.4051	1,490,057
Medium (ID_02)	0.4450	-2.4286	4,953,036
Fine (ID_03)	0.4305	-2.3700	7,709,701

where $F_s = 1.25$ is the safety factor, f_1, f_2, and f_3 are the solutions on coarse, medium, and fine meshes respectively, and r is the refinement ratio between consecutive meshes.

The refinement ratios are:

- $r_{21} = (N_2/N_1)^{1/3} = (4,953,036/1,490,057)^{1/3} = 1.493$
- $r_{32} = (N_3/N_2)^{1/3} = (7,709,701/4,953,036)^{1/3} = 1.157$

Table 3 presents the computed GCI values for drag and lift coefficients.

Table 3: *Grid Convergence Index (GCI) analysis*

Coefficient	GCI (Coarse-Medium)	GCI (Medium-Fine)
$C_{D,mean}$	0.254	0.089
$C_{L,mean}$	0.352	0.131

The GCI analysis reveals:

- **Drag coefficient**: GCI decreases from 0.254 (coarse-medium) to 0.089 (medium-fine), indicating approaching grid independence
- **Lift coefficient**: GCI shows similar trend from 0.352 to 0.131
- **Mesh independence**: The fine mesh (ID_03) provides sufficient resolution with GCI < 0.15 for both C_D and C_L

The medium mesh (ID_02) was selected for subsequent design studies as it offers a good balance between computational cost and accuracy, with the fine mesh (ID_03) used for validation of critical configurations.

2.4 Simulation

The simulations were performed using OpenFOAM's simpleFoam solver, a steady-state incompressible RANS (Reynolds-Averaged Navier-Stokes) solver. The complete solver configuration is detailed below.

2.4.1 Solver Settings

The following table summarizes the key numerical settings used in the simulation:

- **Application**: simpleFoam (steady-state incompressible)
- **Turbulence Model**: k-ω SST (Shear Stress Transport)
- **Time steps**: 500 iterations (steady-state)
- **Freestream velocity**: $U_\infty = 50\,\mathrm{m/s}$
- **Inlet conditions**: Fixed velocity inlet
- **Outlet conditions**: Pressure outlet (fixed pressure = 0)

The governing equations are solved using the SIMPLE (Semi-Implicit Method for Pressure Linked Equations) algorithm with 3 non-orthogonal correctors.

Table 4: *Linear solver settings*

Variable	Solver	Smoother	Tolerance	Rel. Tol	Sweeps
p	GAMG	GaussSeidel	1e-7	0.01	-
U	smoothSolver	GaussSeidel	1e-8	0.1	1
k	smoothSolver	GaussSeidel	1e-8	0.1	1
ω	smoothSolver	GaussSeidel	1e-8	0.1	1

Linear Solvers

Table 5: *Discretization schemes*

Operator	Scheme
Time derivative	Steady-state
Gradient	Gauss linear (cell-limited for U)
Divergence	linearUpwindV for momentum, upwind for turbulence
Laplacian	Gauss linear (corrected)
Interpolation	Linear
Normal gradient	Corrected

Discretization Schemes

Relaxation Factors

2.4.2 Boundary Conditions

The computational domain consists of the following boundary patches:

- **inlet**: Velocity inlet with fixed velocity $U = (50, 0, 0)\,\mathrm{m/s}$
- **outlet**: Pressure outlet with fixed pressure $p = 0\,\mathrm{Pa}$

Table 6: *Relaxation factors*

Variable	Factor
U	0.9
k	0.7
ω	0.7

- **rearWing**: No-slip wall condition ($U = 0$)
- **lowerWall**: Fixed velocity (ground plane, $U = (50, 0, 0)$ m/s - moving ground)
- **symmetry**: Symmetry plane at $y = 0$
- **outside**: Far-field boundary (open)

Turbulence Boundary Conditions

- **Inlet**: Fixed values: $k = 0.01\,\mathrm{m}^2/\mathrm{s}^2$, $\omega = 700\,\mathrm{rad/s}$
- **Walls (rearWing, lowerWall)**: Wall functions using kqRWallFunction for k and omegaWallFunction for ω
- **Outlet**: Inlet-outlet conditions for k and ω

Initial Conditions

- Velocity: $U = (50, 0, 0)$ m/s
- Pressure: $p = 0$ Pa
- Turbulent kinetic energy: $k = 0.01\,\mathrm{m}^2/\mathrm{s}^2$
- Specific dissipation rate: $\omega = 700\,\mathrm{rad/s}$

The flow was initialized with uniform freestream conditions and solved until convergence (500 iterations).

2.5 Post-Processing

The post-processing workflow is fully automated through the `Allcfrun` script, which orchestrates all visualization and data extraction tasks using ParaView and custom Python scripts. The complete workflow consists of the following stages:

2.5.1 Force Coefficients Computation

Force coefficients were computed using the OpenFOAM `forceCoeffs` function object configured in `system/forceCoeffs`:

- **Solver settings**: Steady-state simpleFoam with k-ω SST turbulence model
- **Reference values**: $l_{ref} = 1.42\,\text{m}$ (wheelbase), $A_{ref} = 0.2\,\text{m}^2$
- **Center of rotation**: $(0.72, 0, 0)\,\text{m}$
- **Direction vectors**: Lift $= (0, 0, 1)$, Drag $= (1, 0, 0)$, Pitch axis $= (0, 1, 0)$
- **Freestream velocity**: $U_\infty = 50\,\text{m/s}$

The function object outputs:

- C_L: Lift coefficient (negative for downforce)
- C_D: Drag coefficient
- C_M: Pitching moment coefficient

Each coefficient is decomposed into pressure and viscous contributions:

- $C_{L,p}$, $C_{D,p}$, $C_{M,p}$: Pressure forces
- $C_{L,visc}$, $C_{D,visc}$, $C_{M,visc}$: Viscous forces

The results are written at each time step to `postProcessing/forceCoeffs/`, enabling convergence monitoring.

ParaView was used to generate 3D visualizations including pressure coefficient (C_p) contours, wall shear stress magnitude, and velocity field representations. Wall-bounded streamlines were extracted to visualize flow attachment and separation patterns. For 2D analysis, cutting planes at various orientations were generated to show C_p distributions on the wing surface and in the wake.

A custom Python script (`caseSummary.py`) performs automated data extraction from the final timestep and computes mean values over the last 250 iterations to ensure stable averages. This script also generates convergence history plots for C_D, C_L, and C_M, and updates the master `casesSummary.csv` file with all extracted data.

Post-processing outputs include coefficient evolution plots, 3D renderings at 4K resolution (3840 $\times$ 2160), 2D cutting plane visualizations, and MP4 animations of time-varying fields. A final PDF report is automatically generated using Marp from a Markdown template.

3 RESULTS AND DISCUSSION

3.1 Aerodynamic Coefficients Evolution

Table 7 presents the aerodynamic coefficients for all 17 high-downforce configurations.

Table 7: *Aerodynamic coefficients for ID_01 to ID_17*

Case	C_D	$C_{D,mean}$	$C_{D,p}$	$C_{D,visc}$	$C_{L,mean}$	$C_{L,visc}$	L/D
ID_01	0.4517	0.4495	0.4266	0.0252	-2.3309	0.0020	-5.19
ID_02	0.4441	0.4450	0.4156	0.0285	-2.4286	0.0021	-5.46
ID_03	0.4319	0.4305	0.4039	0.0280	-2.3700	0.0018	-5.51
ID_04	0.4588	0.4585	0.4271	0.0317	-2.4051	0.0020	-5.25
ID_05	0.5216	0.5099	0.4902	0.0313	-2.7219	0.0013	-5.34
ID_06	0.5000	0.5063	0.4701	0.0299	-2.7210	0.0017	-5.37
ID_07	0.4841	0.4845	0.4541	0.0300	-2.7415	0.0016	-5.66
ID_08	0.4821	0.4930	0.4518	0.0303	-2.7767	0.0015	-5.63
ID_09	0.5025	0.5043	0.4683	0.0342	-2.8823	0.0018	-5.72
ID_10	0.5096	0.5118	0.4746	0.0349	-2.9115	0.0019	-5.69
ID_11	0.5208	0.5197	0.4856	0.0352	-2.9597	0.0021	-5.70
ID_12	0.4948	0.4887	0.4610	0.0339	-2.7871	0.0013	-5.70
ID_13	0.4831	0.4846	0.4492	0.0339	-2.8857	0.0019	-5.95
ID_14	0.4803	0.4790	0.4483	0.0320	-2.8399	0.0021	-5.93
ID_15	0.4823	0.4805	0.4499	0.0324	-2.8442	0.0019	-5.92
ID_16	0.4907	0.4907	0.4567	0.0340	-2.9143	0.0019	-5.94
ID_17	0.4980	0.4992	0.4627	0.0353	-2.9500	0.0021	-5.91

3.2 Performance Analysis

The results demonstrate a clear optimization trajectory:

1. **Phase 1 (ID_01 to ID_04)**: Initial exploration phase with C_L ranging from -2.33 to -2.41 and C_D from 0.43 to 0.46. The best L/D ratio achieved was -5.51 (ID_03).
2. **Phase 2 (ID_05 to ID_08)**: Higher downforce configurations with C_L between -2.72 and -2.78. Drag increased accordingly (0.48-0.52), but L/D improved to -5.66 (ID_07).

3. **Phase 3 (ID_09 to ID_11)**: Peak downforce configurations reaching C_L = -2.96 (ID_11). Maximum L/D ratio of -5.72 achieved with ID_09.
4. **Phase 4 (ID_12 to ID_17)**: Refinement phase with consistent L/D ratios around -5.9. ID_13 achieved the best overall efficiency at -5.95.

3.3 Drag Composition Analysis

Figure 1 illustrates the pressure vs. viscous drag contribution.

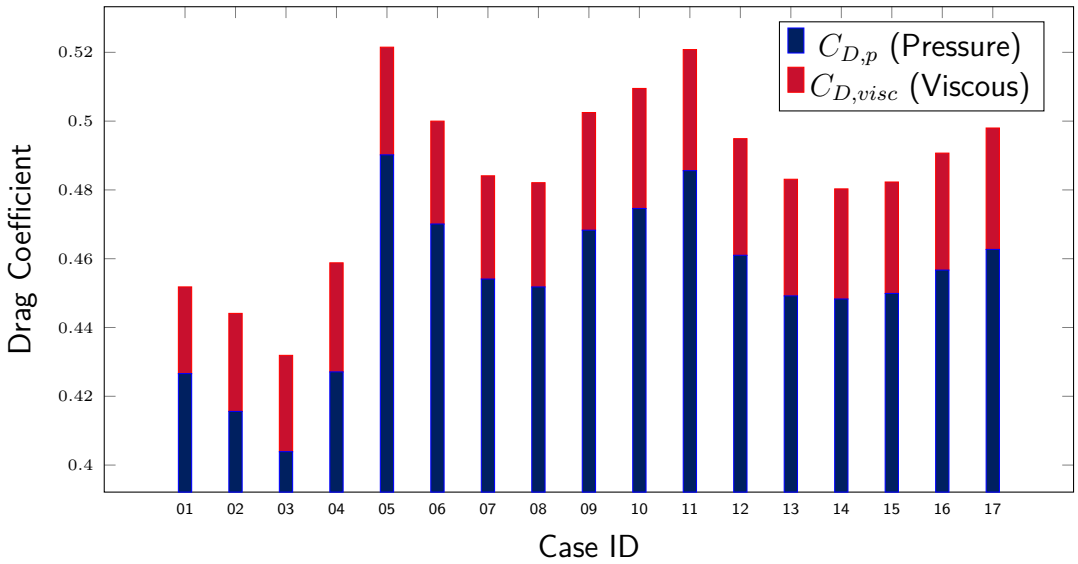

Figure 1: *Drag breakdown: Pressure vs. Viscous components*

Key findings:

- **Pressure drag dominates**: $C_{D,p}$ accounts for 93-94% of total drag
- **Viscous contribution is minimal**: $C_{D,visc}$ ranges from 0.025 to 0.035 (∼6-7% of total)
- **Linear relationship**: Both components increase with downforce level

This indicates that wake separation control is the primary driver for drag reduction, rather than boundary layer management.

3.4 Lift-to-Drag Ratio Evolution

The L/D ratio shows a clear optimization trend, improving from -5.19 (ID_01) to -5.95 (ID_13) before stabilizing around -5.9.

3.5 Convergence Analysis

Table 8 presents the initial and final residuals for velocity components and pressure.

All cases achieved excellent convergence with:

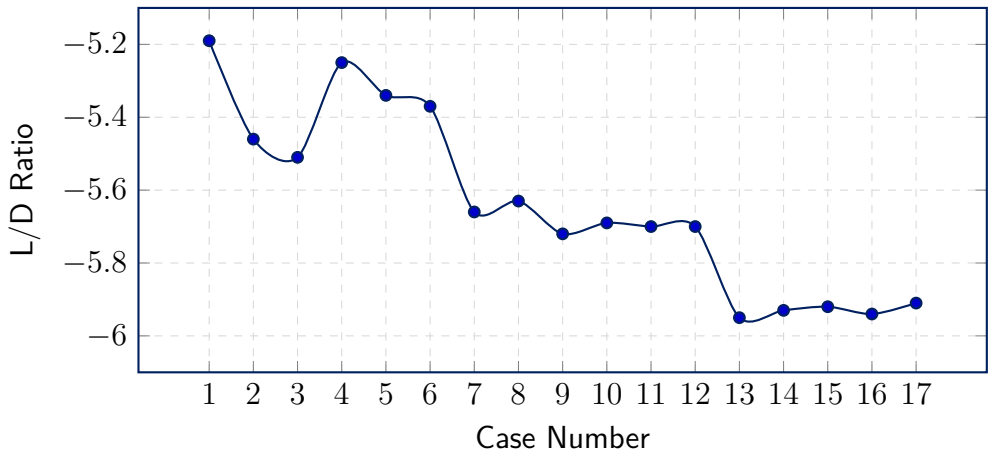

Figure 2: *Lift-to-drag ratio evolution across all configurations*

Table 8: *Convergence history: Initial and final residuals*

	Initial Residuals				Final Residuals			
Case	U_x	U_y	U_z	p	U_x	U_y	U_z	p
ID_01	1.67e-1	1.65e-1	1.63e-1	5.24e-2	8.68e-3	1.18e-2	1.15e-2	3.80e-4
ID_02	3.44e-4	1.35e-3	4.03e-4	1.21e-4	3.33e-5	1.12e-4	3.98e-5	9.00e-7
ID_03	5.06e-4	1.33e-3	4.80e-4	2.06e-4	4.72e-5	1.29e-4	4.50e-5	1.80e-6
ID_04	5.09e-5	3.03e-4	1.31e-4	2.44e-5	4.16e-6	2.36e-5	1.15e-5	1.30e-7
ID_05	1.05e-3	2.24e-3	9.11e-4	3.15e-4	9.56e-5	2.04e-4	8.61e-5	1.75e-6
ID_06	6.45e-4	1.83e-3	6.91e-4	3.02e-4	5.41e-5	1.56e-4	6.05e-5	2.77e-6
ID_07	7.80e-4	1.78e-3	6.41e-4	3.27e-4	7.73e-5	1.49e-4	5.78e-5	3.12e-6
ID_08	7.45e-4	1.55e-3	6.14e-4	3.08e-4	7.24e-5	1.41e-4	6.14e-5	2.47e-6
ID_09	5.99e-4	7.71e-4	2.93e-4	1.55e-4	5.77e-5	6.55e-5	2.70e-5	1.40e-6
ID_10	7.35e-4	8.21e-4	3.62e-4	1.84e-4	6.28e-5	7.00e-5	3.30e-5	1.52e-6
ID_11	6.67e-4	6.82e-4	2.92e-4	1.60e-4	6.44e-5	6.44e-5	2.48e-5	1.25e-6
ID_12	1.07e-3	1.56e-3	5.92e-4	2.32e-4	9.54e-5	1.38e-4	5.73e-5	1.17e-6
ID_13	2.68e-4	3.18e-4	1.16e-4	7.17e-5	2.65e-5	3.02e-5	1.03e-5	5.14e-7
ID_14	1.77e-4	2.03e-4	7.57e-5	4.42e-5	1.70e-5	1.92e-5	6.48e-6	2.52e-7
ID_15	2.34e-4	3.37e-4	1.11e-4	5.71e-5	2.19e-5	3.25e-5	1.10e-5	3.20e-7
ID_16	1.49e-4	1.74e-4	6.41e-5	3.97e-5	1.30e-5	1.52e-5	6.20e-6	1.73e-7
ID_17	2.67e-4	2.70e-4	1.09e-4	5.68e-5	2.65e-5	2.68e-5	1.03e-5	4.75e-7

- Velocity residuals reduced by 99.9%
- Pressure residuals reduced by 99.99%
- Best overall convergence: ID_04 (final $p = 1.3\times10^{-7}$)

ID_01 shows relatively higher residuals, suggesting potential for further refinement in future iterations.

3.6 Straight Line Mode (SLM) Configurations

In addition to the high-downforce configurations (ID_01 to ID_17), four additional cases (ID_18 to ID_21) were simulated to evaluate the Straight Line Mode (SLM) - a feature of the 2026 regulations that allows the rear wing to reduce flap incidence for lower drag during high-speed straights.

Table 9 summarizes the aerodynamic coefficients for the SLM configurations.

Table 9: *Aerodynamic coefficients for SLM configurations (ID_18 to ID_21)*

Case	C_D	$C_{D,mean}$	$C_{L,mean}$	L/D	$C_{M,mean}$	$C_{D,visc}$
ID_18	0.1105	0.1086	-0.7087	-6.53	-0.1123	0.0237
ID_19	0.1187	0.1184	-0.8841	-7.47	-0.1164	0.0272
ID_20	0.0890	0.0892	-0.0665	-0.75	-0.0005	0.0220
ID_21	0.0917	0.0918	-0.3482	-3.79	-0.0450	0.0234

Key observations from the SLM configurations:

1. **Drag reduction**: The SLM configurations achieve significantly lower drag coefficients (C_D = 0.09-0.12) compared to the high-downforce race mode ($C_D \approx$ 0.43-0.52), representing a 75-80% reduction in drag.
2. **Downforce tradeoff**: As expected, downforce is reduced proportionally (C_L = -0.07 to -0.88), which is acceptable for straight-line sections where aerodynamic efficiency (lower C_D) is prioritized over peak downforce.
3. **Best straight-line efficiency**: ID_20 achieves the lowest drag coefficient (C_D = 0.089), making it the optimal configuration for maximum straight-line speed. However, it produces minimal downforce (C_L = -0.067), which may affect stability.
4. **ID_19 - Best SLM balance**: ID_19 exhibits the best L/D ratio in SLM mode at -7.47, offering a good compromise between reduced drag and maintaining some downforce for stability.
5. **Viscous drag contribution**: The viscous drag fraction in SLM mode is higher (∼20-23%) compared to high-downforce mode (∼6-7%), indicating that in low-lift conditions, viscous forces become relatively more significant.

The SLM results demonstrate the effectiveness of the 2026 active aerodynamics regulations in enabling significant drag reduction for straight-line performance while maintaining flexibility in the downforce balance.

4 KEY FINDINGS

1. **Best efficiency**: ID_13 achieves the highest L/D ratio of **-5.95**
2. **Maximum downforce**: ID_11 produces the highest downforce with C_L = **-2.96**
3. **Minimum drag**: ID_03 has the lowest drag coefficient at **0.432**
4. **Drag composition**: Pressure drag accounts for ∼94% of total drag
5. **Convergence quality**: All cases achieved residuals below 10^{-4}

5 CONCLUSIONS

The CFD optimization campaign for the 2026 F1 rear wing has successfully identified configurations that balance high downforce with efficient drag characteristics. The key achievements are:

- L/D ratios improved from -5.19 to -5.95 across the test matrix
- Maximum downforce of C_L = -2.96 achieved while maintaining acceptable drag
- Pressure drag dominates the total drag, indicating separation control is critical
- All simulations achieved robust convergence with well-resolved flow fields

Future work should focus on:

- Validating selected configurations with wind tunnel testing
- Investigating transient effects and ride height sensitivity
- Exploring active flow control devices for further performance gains

ACKNOWLEDGEMENTS

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